Government Engineering College Kishanganj

Department of Computer Science and Engineering

Computer Networks LAB-01

Experiment Title

Study of LAN Topology, Network Devices, and Address Configuration

Problem Statement

Perform the following tasks and prepare a detailed report:

1. **Draw the Network Topology:**
    - Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
    - Clearly label all components such as Router, Switch, Access Point, and end devices (PCs, laptops, etc.).
2. **Identify Network Devices:**
    - List all networking devices used in the LAN.
    - Mention the function of each device.
3. **Analyze Connection Types:**
    - Specify whether connections are:
        - Wired (Ethernet)
        - Wireless (Wi-Fi)
    - Mention where each type is used.
4. **Measure Network Speed:**
    - Find the link speed of:
        - Wired network (if available)
        - Wireless network
    - Mention the tools/commands used (e.g., Task Manager, `netsh; netsh wlan show interfaces or` , etc.).
5. **Find System Network Details:**
    - Determine and record the following of your computer:
        - MAC Address
        - IP Address (IPv4)

    - Subnet Mask
  - Mention the command(s) used (e.g., `ipconfig`, `getmac`).

6. **Explain Observations:**
  - Briefly explain:
    - Difference between wired and wireless speed
    - Role of subnet mask in networking

Output

- Network diagram (hand-drawn not software-generated)
- List of devices and their roles
- Screenshots/outputs of commands used
- Table of network parameters (MAC, IP, Subnet Mask)
- Short explanation and conclusion